MILITARY SPECIFICATION

BEAKER-FLASK, Laboratory

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers beaker-flasks for general laboratory use.

1.2 Classification. Beaker-flasks covered by this specification shall be of the following sizes (see 6.1):

Sizes

1 - 150 milliliters (ml.)
2 - 300 ml.
3 - 500 ml.
4 - 800 ml.
5 - 1000 ml.
6 - 1200 ml.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

DD-G-541	Glass (Laboratory).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

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Military

MIL-P-116	Preservation-packaging, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM C148	Polariscopic Examination of Glass Containers.
ASTM C162	Definitions of Terms Relating to Glass and Glass Products.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as herein specified.

3.1.1 Glass. Glass shall conform to Type I, Class A borosilicate glass specified in DD-G-541.

3.1.2 Plastic. Plastic shall be black polypropylene.

3.1.3 Rubber. Rubber shall be neutral (or resin-cured butyl) rubber.

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3.2 Design and construction. The beaker-flask shall consist of a glass container with a plastic cap containing a rubber insert. Figure 1 shows the general appearance of the beaker-flask. The glass container, when capped, shall not leak when tested as specified in 4.4.1. The capped container shall be stable on a flat surface.

3.2.1 Glass container. The glass container shall be symmetrically fabricated (except for the spout) and smoothly finished with straight sides with at least a 3/8 inch outside radius at the junction with the bottom surface. The neck shall be flared outward at the top to form the mouth which shall have a fire-polished slightly thickened edge with a pouring spout. The bottom surface shall be flat or slightly concave.

3.2.1.1 Dimensions. The glass containers shall conform to the requirements of Table I.

3.2.1.1.1 Capacity fill. The capacity fill of the container shall be as specified in Table I when tested as indicated in 4.4.2.for sizes 1, 2 and 3.

3.2.1.2 Accuracy. The container shall have a volumetric accuracy of ± 5% when tested as indicated in 4.4.3.

Table I Glass container dimensions *

Size	Capacity (ml.)	Height (in.)	O.D. Base (in.)	O.D. of mouth at spout (in.)
1	150	3.1 (7.87 cm.)	2.75 (6.99 cm.)	2.38 (6.05 cm.)
2	300	4.9 (12.45 cm.)	2.75 (6.99 cm.)	2.38 (6.05 cm.)
3	500	7.0 (17.78 cm.)	2.75 (6.99 cm.)	2.38 (6.05 cm.)
4	800	7.5 (19.05 cm.)	3.75 (9.53 cm.)	2.84 (7.22 cm.)
5	1000	8.75 (22.23 cm.)	3.75 (9.53 cm.)	2.84 (7.22 cm.)
6	1200	10.0 (25.4 cm.)	3.75 (9.53 cm.)	2.84 (7.22 cm.)

* All dimensions shall have a tolerance of ± 0.1 inch (± .25 cm.).

3.2.2 Cap. The cap shall be of the style and design shown in Figure 1. The cap shall be capable of being cleaned with soap and warm water. The cap shall not warp, crack or deteriorate when autoclaved as specified in 4.4 4.

3.2.2.1 Dimensions. The caps shall comply with the requirements of Table II. Cap shall fit the container securely, and container shall not leak (see 3.2).

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Table II Cap dimensions *

Bottle size	O.D. (in.)	Overall height
1	2.75 (6.99 cm.)	1.65 (4.22 cm.)
2	2.75 (6.99 cm.)	1.65 (4.22 cm.)
3	2.75 (6.99 cm.)	1.65 (4.22 cm.)
4	3.75 (9.53 cm.)	2.35 (6.02 cm.)
5	3.75 (9.53 cm.)	2.35 (6.02 cm.)
6	3.75 (9.53 cm.)	2.35 (6.02 cm.)

* All dimensions shall have a tolerance of ± 0.1 inch (±.25 cm.).

3.3 Graduation markings. Graduation markings shall be sharply defined lines of uniform width, perpendicular to the vertical axis of the container and parallel to each other. Graduation markings, capacity and accuracy markings shall be applied in a legible and permanent manner to the external surface of the glass container. Marking shall be applied in enamel by the silk-screen process. Aforementioned markings shall not lose legibility when tested as specified in 4.4.5. Graduation markings and numerical capacity values required for the various sizes of beaker-flasks shall be as shown on Table III.

Table III Graduation markings

Size	Graduation markings
1	40 to 120 ml. in 20 ml. increments with 40, 80 and 120 ml. numerical values marked.
2	50 to 250 ml. in 50 ml. increments with 50, 150 and 250 ml. numerical values marked.
3	50 to 450 ml. in 50 ml. increments with each 100 ml. capacity numerical value marked.
4	50 to 800 ml. in 50 ml. increments with each 100 ml. capacity numerical value marked.
5	50 to 1000 ml. in 50 ml. increments with each 100 ml. capacity numerical value marked.
6	50 to 1200 ml. in 50 ml. increments with each 100 ml. capacity numerical value marked.

3.4 Area for marking. Each beaker-flask shall have an area for marking. Marking area shall be suitable for marking after permanency of marking test specified in 4.4.5.

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4.5 Annealing. Ring strain shall be no greater than Temper No. 1 when the glass container is tested as specified in 4.4.6. Any longitudinal strain, if present, shall be faint and highly diffused.

3.6 Identification marking. Each beaker-flask shall be permanently marked with the manufacturer's name or trademark on the glass container.

3.7 Workmanship. Glass containers shall be transparent, free of striae, surface irregularities and other imperfections which cause distortion of light through the glass. The container shall be free of blisters, boils, seeds, stones, chips, cracks or other defects which may weaken the glass or impair the serviceability of the container. Cap shall be free of cracks, tears, holes or other imperfections which affect appearance or serviceability of the cap.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificate of quality. Certificate of quality, supplied by the manufacturer of the glass, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the item identification, the name of the component, the lot number, the lot size, the sample size

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the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table IV.

Table IV Sampling for examination *

Type of defect	Inspection level	AQL (percent defective)
For visual examination		
Major defects	II	1.0
Minor defects	II	2.5
For dimensional examination	S-2	2.5

* Unit of product shall be the beaker-flask.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table V.

Table V Sampling for tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Inspection level	AQL
Beaker-flask	Leakage	3.2	4.4.1	S-1	*
Glass container	Capacity fill	3.2.1.1.1	4.4.2	S-1	*
Glass container	Accuracy	3.2.1.2	4.4.3	S-1	*
Cap	Autoclaving	3.2.2	4.4.4	S-1	*
Glass container	Permanency of marking	3.3 or 3.4	4.4.5	S-1	*
Glass container	Annealing	3.5	4.4.6	S-1	*

4.3 Examination. The beaker-flask shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following table:

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Table VI Classification of defects *

Categories	Defects
Major	
101	Container not free of chips, cracks and breaks.
102	Container not free of checks, blisters and stones.
103	Glass not clear and transparent.
104	Container not free of cracks and sharp edges.
105	Lip of container not fire-polished.
106	Beaker-flask not stable on flat surface.
107	Junction of sides and bottom not well-rounded.
108	Cap not free of cracks and sharp edges.
109	Cap insert torn or not seated properly.
Minor	
201	Container not free of scratches, seeds and striae.
202	Container not free of rough edges, protrusions and deformed areas.
203	Container not free of thin areas.
204	Beaker-flask not free of grease, oil and foreign matter.
205	Identification marking incomplete or illegible.

* Examination shall not be restricted to the classified possible defects listed above. Glass defects are as defined in ASTM Designation C162.

4.3.2 Dimensional examination. The beaker-flask shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.3.3 Packaging inspection. The inspection of the preservation-packaging and interior package marking shall be in accordance with the group A and B quality conformance inspection requirements of MIL-P-116. The inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Leak test. The beaker-flask shall be filled with colored water, normally capped; inverted and placed cap down on a paper towel at room temperature for a period of 30 minutes. Leakage shall be evidenced by liquid in the cap or color on the paper towel. (Room temperature shall be 25°C. ± 5°.)

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4.4.2 Capacity fill. A volume of distilled water equal to the capacity specified for the size shall be added to the container. The water level shall be between the shoulder and the neck of the container. The test shall be conducted at room temperature (25 ± 5°C.).

4.4.3 Accuracy. Container shall be filled with distilled water to the top numerically-marked graduation mark. The volume of water used shall be determined and recorded. The test shall be conducted at room temperature (25 ± 5°C.).

4.4.4 Autoclaving. The cap shall be autoclaved at 121°C. for 15 minutes.

4.4.5 Permanency of marking. A different beaker-flask shall be used for each of the following tests:

4.4.5.1 Resistance to alkali. The glass container shall be completely immersed in 1N sodium hydroxide solution at room temperature, covered, boiled for 30 minutes and allowed to stand in the alkali for an additional 60 minutes. The glass container shall then be removed from the solution, thoroughly rinsed with distilled water and dried by rubbing vigorously with a cloth.

4.4.5.2 Resistance to acid. The glass container shall be completely immersed in concentrated hydrochloric acid (specific gravity 1.18) at room temperature, covered, boiled for 30 minutes, and allowed to stand in the acid for an additional 90 minutes. The glass container shall be removed from the solution, thoroughly rinsed with distilled water and dried by rubbing vigorously with a cloth.

4.4.6 Annealing. Glass container shall be tested in accordance with method A or B of ASTM C148.

4.4.7 Packaging inspection. The inspection of the unit, intermediate package, packing and marking for shipment and storage shall be in accordance with quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

5. PACKAGING

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package, containing 10 beakers with caps of size 1, 2 or 3 or one package containing 5 beakers with caps of size 4, 5 or 6, as specified, constitutes one unit of issue.

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5.1.1.1 Level A.

5.1.1.1.1 Unit packaging. Each unit of issue shall be packaged in accordance with MIL-P-116, method III. Box shall conform to PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. The number of units specified in Table VII shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed as specified in the appendix of PPP-B-636.

5.2.2 Level B. The number of units specified in Table VII shall be packaged in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

Table VII Exterior pack quantity

Size	Quantity
1	10 units
2	6 units
3, 4 of 5	4 units
6	2 units

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking. Each unit package and exterior container shall be marked as specified in MIL-STD-129. In addition, the exterior container shall be marked with the legend:

"GLASS--HANDLE WITH CARE."

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Size required (see 1.2)
d. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification covers the following items listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-01-013-3553	BEAKER-FLASK, Graduated, 150 ml.
6640-01-013-5280	BEAKER-FLASK, Graduated, 300 ml.
6640-01-013-3551	BEAKER-FLASK, Graduated, 500 ml.
6640-01-013-3555	BEAKER-FLASK, Graduated, 800 ml.
6640-01-013-3552	BEAKER-FLASK, Graduated, 1000 ml.
6640-01-013-3554	BEAKER-FLASK, Graduated, 1200 ml.

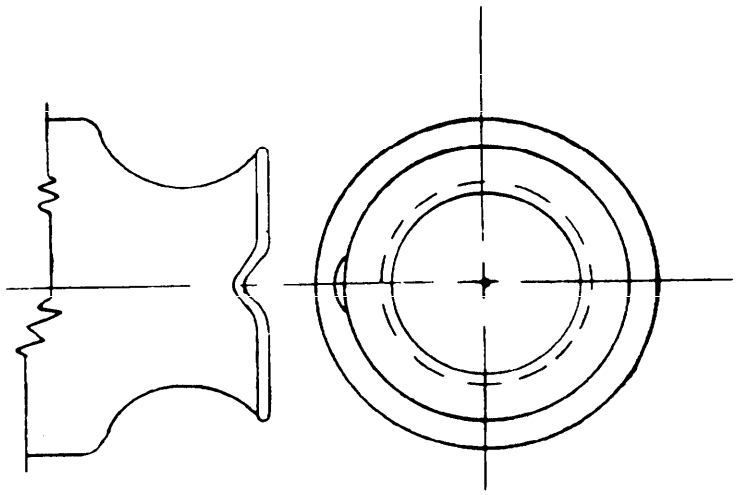
6.3 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

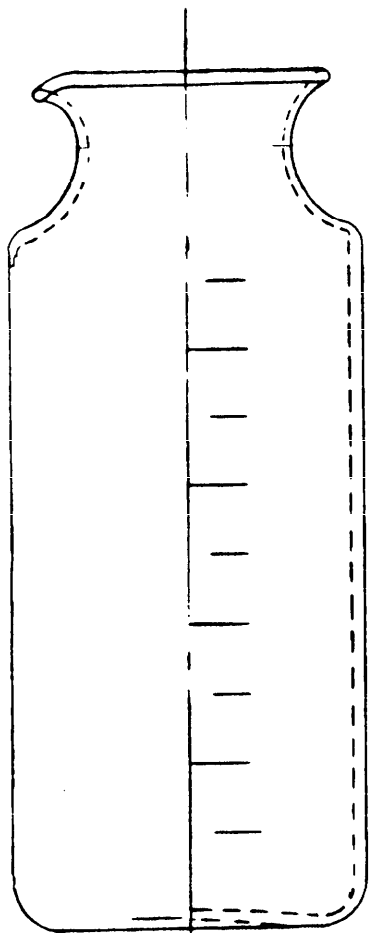
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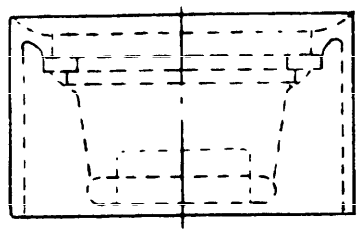
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BOTTLE



CAP

FIGURE 1 - BEAKER-FLASK, Graduated

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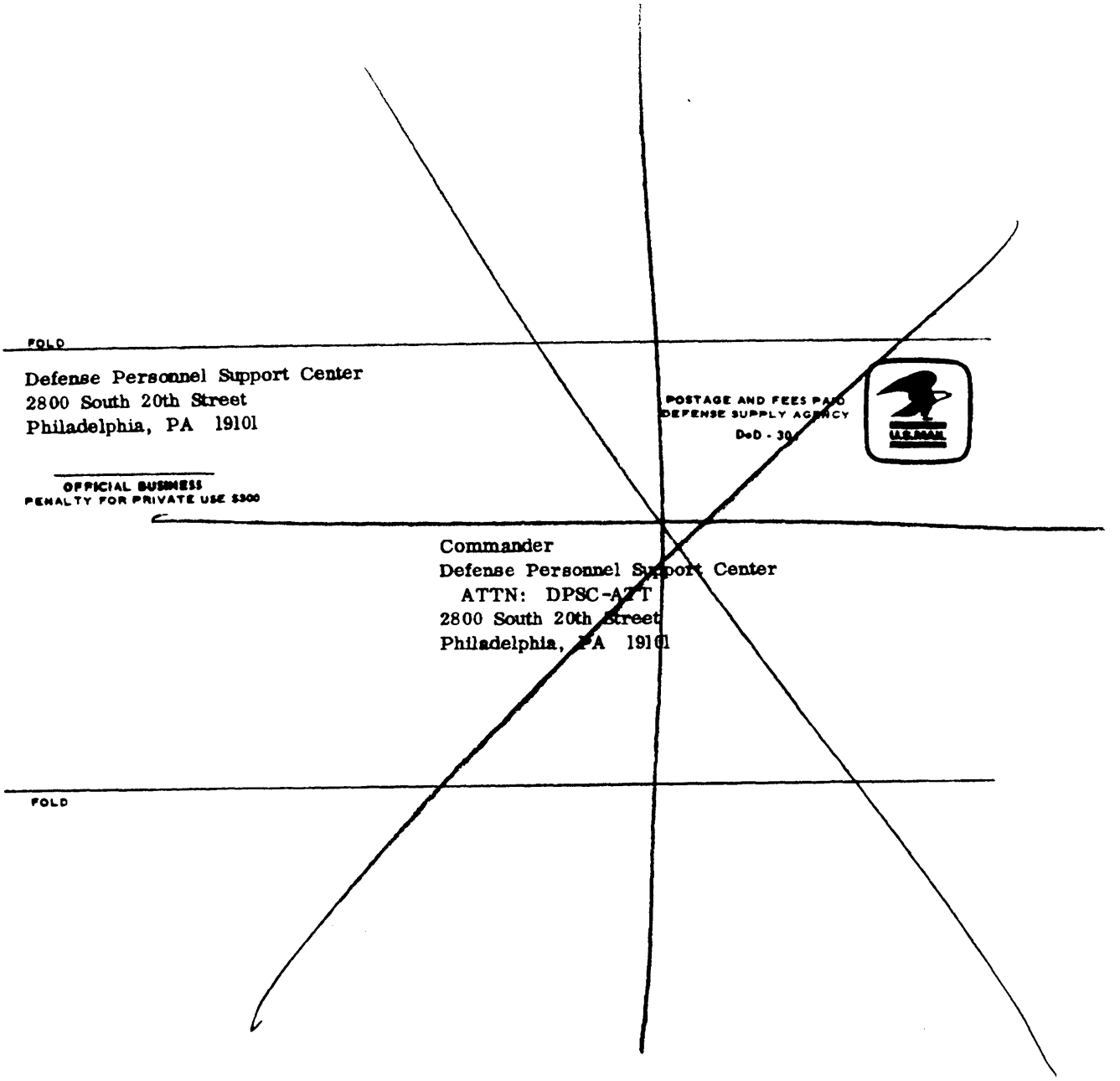
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